WashU McKelvey Engineering

Spring 2026 MEMS 3430 Design of Thermal Systems

Design Homework III - 2nd Submission

CCGT – Heat Exchanger Design

Assumptions and Boundary Conditions

Assumptions.

(1) Each heat exchanger is perfectly insulated, such that $q_c = q_h$. A real system would include losses to the environment in the heat exchanger, but that is neglected for simplicity [1].
(2) Axial conduction along the tubes is negligible due to thin tube walls. A real system would include losses along the tubes, but that is neglected for simplicity [1].
(3) Potential and kinetic energy changes are negligible, so there are negligible height or fluid velocity changes throughout the heat exchangers [1]. This allows for ease of calculation, but in reality the specific geometries of the heat exchanger would affect the potential and kinetic energies.
(4) Thermophysical properties can be determined at a representative temperature and assumed to be constant for small changes in fluid temperatures. This is a simplification that allows for the properties of the working fluids to be generalized.
(5) The system is assumed to be operating at steady-state, and does not take into account any start-up or shut-down behavior that would change results.
(6) Unless otherwise specified, system components in the Brayton and Rankine cycle are assumed to be ideal and have perfect efficiency.
(7) The applicable ranges of Reynolds numbers in the turbulent regime were neglected for the sake of performing simpler calculations, reducing the accuracy of some $Nu - Re$ correlations.
(8)

Boundary Conditions.

(1) The heat transfer between the topping and bottom cycles is 30 MW.
(2) The isentropic efficiency of the Rankine turbine is 90%.
(3) The isentropic efficiency of the Brayton cycle turbine is 91%, and the isentropic efficiency of the Brayton cycle compressor is 87% [2].
(4) The pressure in state 6 is equal to the pressure in state 7, and is equal to 310 KPa [3].
(5) The temperature in state 7 is 525.85 K, and the enthalpy is 2973.02 kJ/kg [3].
(6) The temperature in state 6 is 308.49 K, and the enthalpy is 148.361 kJ/kg [3].
(7) The temperature in state 5 is 500 K [2].
(8) The temperature in state 4 is 533.85 K [3].
(9) The mass flow rate of the Brayton cycle is 882.2 kg/s [2], and the mass flow rate of the Rankine cycle is 10.62 kg/s [3].
(10) The temperatures of the fluid in the topping cycle must remain at higher temperatures than that of the bottoming cycle in order for heat to transfer, such that $T_5 > T_6$, and $T_4 > T_7$.
(11) The baffle spacing is taken to be 50% of the inner diameter of the shell-and-tube heat exchanger [4].
(12) The tube-to-tube pitch distance is taken to be 1.25 times a tubes outer diameter [5].
(13) The tube layout is taken to be a triangular pitch with a 30 degree layout [4].
(14) There are 200 thin-walled tubes in each heat exchanger, with the diameter of the preheater tubes

being 20mm, while the evaporator and superheater tube diameters are 50mm. The outer tube of the concentric tube in the superheater is 80mm.

Schematics and Diagrams

Figure 1 shows the schematic of the Combined Cycle Gas Turbine (CCGT) featuring the state-numbering utilized throughout the design homework.

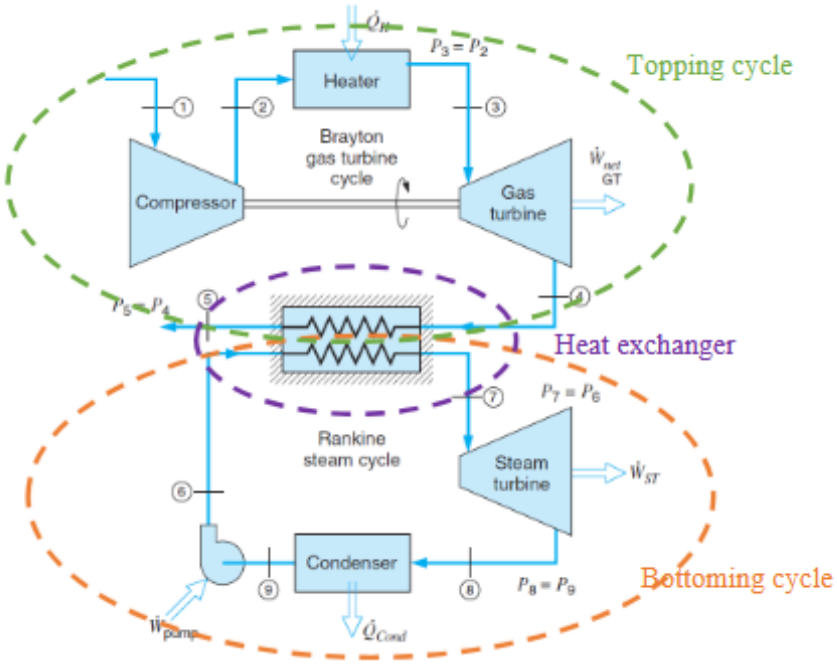


Figure 1 Schematic of CCGT used for the design homeworks

The CCGT consists of a Brayton topping cycle that uses air as the working fluid, a Rankine bottoming cycle that uses water as the working fluid, and a heat exchanger that transfers 30 MW of thermal energy from the exhaust of the topping cycle to the bottoming cycle. In DHW I, the Brayton cycle was optimized to maximize the overall efficiency and net power output of the cycle. In DHW II, the Rankine cycle was optimized to maximize the amount of useful work that could be produced by the cycle. In this DHW, the goal is the determine the required lengths of tubes in the heat exchanger. Rather than treat the heat exchanger as one unit, it was broken down into three seperate systems, shown in Fig. 2.

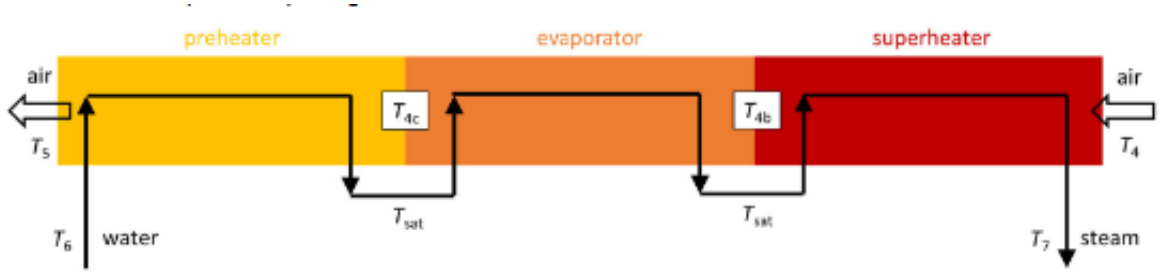


Figure 2 Schematic of the three systems comprising the heat exchanger sections

The preheater section is where the bottoming cycle's liquid water at state 6 is heated to saturation temperature utilizing a shell-and-tube heat exchanger with a single shell and two tube passes. The air traveling through the system goes from state 4,c to state 5, which is exhausted to the environment. The middle section is the evaporator, which is a fire-tube boiler that causes water from the Rankine cycle to boil on the outside of tubes in which hot air from the Brayton cycle flows. In the superheater section, the boiled steam is then heated to the turbine inlet temperature T_7 by a counterflow heat exchanger with two concentric tubes. Figure 3 shows the T-x diagram for the preheater heat exchanger.

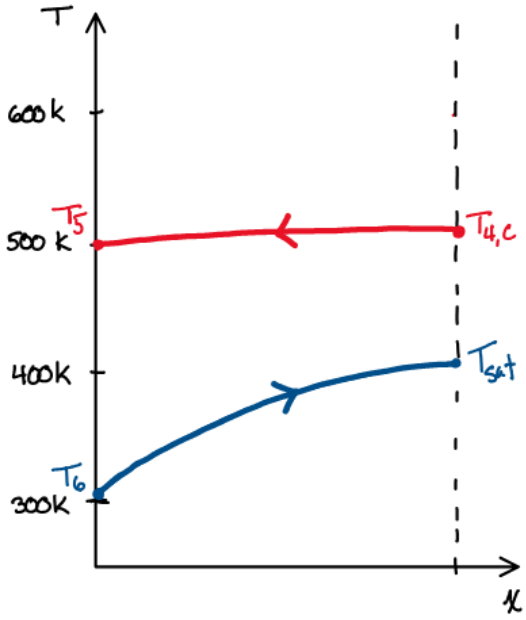


Figure 3 T-x Diagram of the Preheater Heat Exchanger

The preheater operates in counterflow, so the directions of temperature change are drawn in opposite directions. The water enters the cycle at state 6 and is heated by the hot exhaust to saturation, while the topping fluid experiences very little temperature change throughout the process. Fig. 4 shows the T-x diagram for the evaporator heat exchanger.

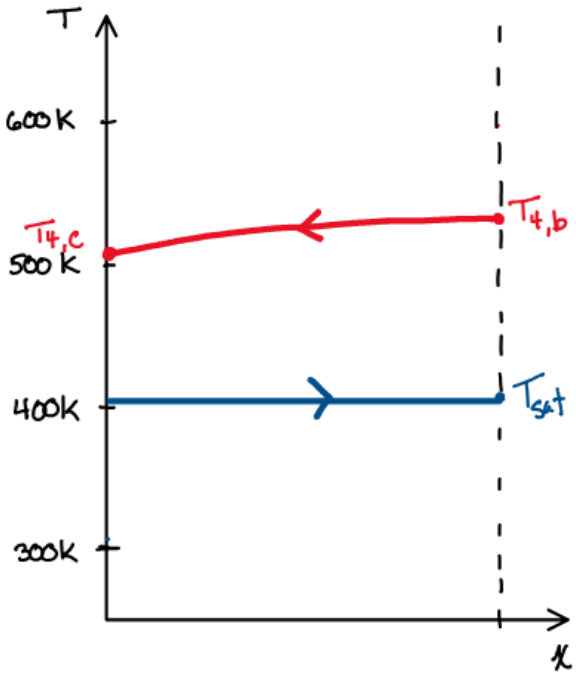


Figure 4 T-x Diagram of the Evaporator Heat Exchanger

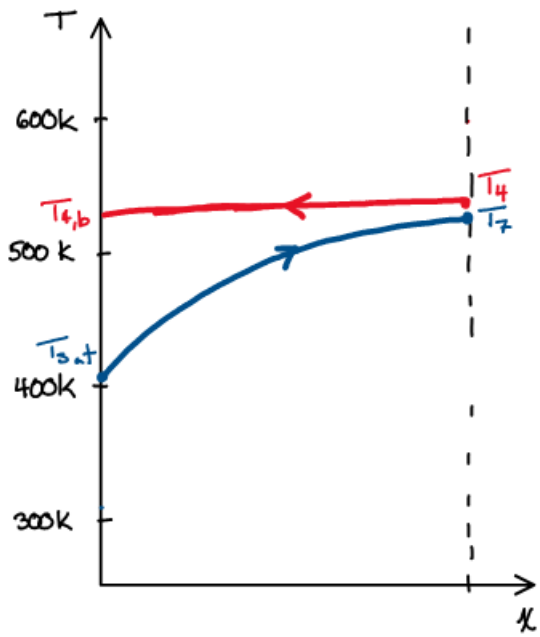


Figure 5 T-x Diagram of the Superheater Heat Exchanger

Equations and Calculations

The first step in the design process was to first calculate the heat transfer rate in each of the three components of the heat exchanger. This process was done using energy balances on each section, with known values from previous design homeworks. Beginning with the preheater section, it is assumed that $q_h = q_c$, so the energy balance can be split in half, such that:

$$Q_{PH} = \dot{m}_{\text{rankine}} (h_f - h_6)$$

Where Q_{PH} is the heat transfer rate from the Brayton cycle to the Rankine cycle in kW, $\dot{m}_{rankine}$ is the mass flow rate of the Rankine cycle from design homework II, 10.62 kg/s. h_f is the enthalpy of saturated water, and h_6 is the enthalpy of state 6, both in kJ/kg. From design homework II, the pressure across the heat exchanger is known to be 301 kPa, so linear interpolation for the enthalpy of saturated water at 301 kPa finds that h_f = 566.16 kJ/kg. The enthalpy of state 6 is known from design homework II to be 148.361 kJ/kg. Solving for Q_{PH},

$$\begin{aligned} Q_{PH} &= \dot{m}_{\text{rankine}} (h_f - h_6) \\ &= (10.62)(566.16 - 148.361) \\ &= \boxed{4437.03 \text{ kW}} \end{aligned}$$

Similarly for the evaporator, the energy balance can be split between the Brayton and Rankine cycle, such that the enthalpy difference across the evaporator is equal to h_{fg}, the difference in enthalpy of water to change from saturated water to saturated vapor. Using linear interpolation at P_6, the enthalpy of saturated vapor is 2726.78 kJ/kg. Solving for Q_{EV},

$$\begin{aligned} Q_{EV} &= \dot{m}_{\text{rankine}} h_{fg} \\ &= 10.62(2726.78 - 566.16) \\ &= \boxed{22946.6 \text{ kW}} \end{aligned}$$

Again, the heat transfer rate in the superheater can be found, using the enthalpy of the saturated vapor previously found, and the enthalpy of state 7 found in design homework II to be 2973.02 kJ/kg. Solving for Q_{SH},

$$\begin{aligned} Q_{SH} &= \dot{m}(h_7 - h_g) \\ &= 10.62(2973.02 - 2726.78) \\ &= \boxed{2615.9 \text{ kW}} \end{aligned}$$

It is known that the total heat transfer rate from the Brayton to the Rankine cycle is 30 MW, so the final step is the check that the combination of the heat transfer rates from the three components equal this total:

$$Q_{total} = 4437 + 22946.6 + 2615.9 = 29,999.5 \text{ kW} \approx \boxed{30 \text{ MW}}$$

The small difference between the expected versus the calculated overall heat transfer rate can be attributed to rounding made during calculations, which would propagate error through all calculations and result in a result that was not exactly 30 MW, but will be considered acceptably close for the purposes of this assignment.

Now that the heat transfer rates were determined, the same energy balances could be applied to the Brayton cycle to find the internal inlet temperatures of each component, using the fact that for small temperature changes $h \approx c_p \Delta T$. Rearranging the energy balances, it is possible to solve for the two inlet temperatures, $T_{4,b}$ and $T_{4,c}$, with $\dot{m}_{brayton}$ being the mass flow rate of the Brayton cycle from design homework I in kg/s, and the $c_{p,air}$ being the specific heat of air at a representative temperature of T_4 for the first calculation, and $T_{4,b}$ for the second calculation.

$$\begin{aligned} T_{4,b} &= T_4 - \frac{Q_{SH}}{\dot{m}_{brayton} c_{p,air}} \\ &= 533.85 - \frac{2615.9}{(882.2)(1.03677)} \\ &= \boxed{530.99 \text{ K}} \end{aligned}$$

$$\begin{aligned} T_{4,c} &= T_{4,b} - \frac{Q_{EV}}{\dot{m}_{brayton} c_{p,air}} \\ &= 530.99 - \frac{22946.6}{(882.2)(1.0362)} \\ &= \boxed{505.89 \text{ K}} \end{aligned}$$

The saturation temperature inside the heat exchanger, T_{sat}, is the saturation temperature of water at P_6, 407.8 K.

With this information, it is now possible to break the system into components to determine the length of the tubes.

Preheater. The preheater is a shell-and-tube heat exchanger with a single shell and two tube passes, with water inside tubes move from state 6 to saturation, and air from the Brayton cycle traveling from the evaporator to the environment. To find the length of the tubes, a $\varepsilon - NTU$ process was followed, beginning with the calculations of C_{min} and C_{max}. Generally, the heat capacity rate C is defined as:

$$C = \dot{m} c_p$$

Which is in units of W/K. The specific heat of the air, $C_{p,a}$, was found using a mean temperature of the air between $T_{4,c}$ and T_5, at a pressure of 1 atm, to be 1030.433 $\frac{\text{J}}{\text{kg} \cdot \text{K}}$. The specific heat of the water,

$C_{p,w}$ was found using a mean temperature of the water between T_6 and T_{sat}, at a pressure of 310 kPa, to be 4200.297 $\frac{\text{J}}{\text{kg·K}}$. Calculating the respective heat capacity rates:

$$C_w = \dot{m}_{rankine} c_{p,w} = 10.62 \cdot 4200.297 = 44,606.967 \text{ W/K}$$

$$C_a = \dot{m}_{brayton} c_{p,a} = 882.2 \cdot 1030.433 = 909,047.653 \text{ W/K}$$

Therefore, $C_{min} = C_w$ and $C_{max} = C_a$. C_r, or the heat capacity rate ratio, can be found using

$$C_r = \frac{C_{min}}{C_{max}} = \frac{44,606.967}{909,047.653} = 0.0491$$

It is next necessary to find Q_{max}, the maximum possible heat transfer rate in W, using the relation

$$Q_{max} = C_{min}(T_{h,i} - T_{c,i})$$

Where $T_{h,i}$ is the temperature of the hot air entering the preheater, $T_{4,c}$, and $T_{c,i}$ is the temperature of the cold water entering the preheater. These two temperatures are used as they represent the largest temperature difference in the system, and C_{min} is used as the limiting factor controlling total heat transfer. Inputting the values,

$$Q_{max} = 44,606.967(505.89 - 308.49) = 8805.415 \text{ kW}$$

The effectiveness of a heat exchanger is defined as $\varepsilon = \frac{Q}{Q_{max}}$, which are now both known for the preheater:

$$\varepsilon = \frac{Q_{PH}}{Q_{max}} = \frac{4437.03}{8805.415} = 0.5039$$

The effectiveness and heat capacity rate ratio can then be used to find the number of heat trasnfer units, NTU. The equation for NTU in a shell-and-tube heat exchanger with 2 tube passes is defined as

$$NTU = -(1 + C_r^2)^{-1/2} \ln(\frac{E - 1}{E + 1})$$

Where E is

$$E = \frac{2/\varepsilon - (1 + C_r)}{(1 + C_r^2)^{1/2}} = \frac{\frac{2}{0.5039} - (1 + 0.0491)}{(1 + 0.0491^2)^{1/2}} = 2.916$$

and the NTU is then

$$NTU = -(1 + 0.0491^2)^{-1/2} \ln(\frac{2.916 - 1}{2.916 + 1}) = 0.7138$$

Now that NTU is known, UA, or the overall heat transfer coefficient multiplied by the total heat transfer surface area, can be found in Eq. 1:

$$UA = NTU \cdot C_{min} = 0.7138 \cdot 44606.967 = 31842.118 \text{ W/K} \quad (1)$$

To find the length, U must first be found, so a $Re - Nu$ approach is selected. The Reynolds number

for water in a circular tube can be found using

$$Re_w = \frac{4\dot{m}}{\pi d \mu_w}$$

Where $\dot{m}$ is the mass flow rate of a single tube, $\frac{\dot{m_w}}{N}$, where N is the number of tubes, d is the diameter of the tube in meters, and μ_w is the dynamic viscosity of water, at the mean water temperature and P_6. Inputting values:

$$Re_w = \frac{4(\frac{10.62}{200})}{\pi(0.020)(0.00033315)} = 10,146.88$$

Turbulent flow inside a pipe is defined as $Re > 10,000$, and Eq. 2 is the Nusselt correlation for turbulent flow:

$$Nu = 0.023 Re^{0.8} Pr^n \tag{2}$$

Where Pr is the Prandtl number of the water, found to be 2.088 at a mean temperature and P_6. n is equal to 0.4 when the fluid is being heated [**?**]. Inputting values:

$$Nu = 0.023(10,146.88)^{0.8}(2.088)^{0.3} = 49.51$$

The Nusselt number is defined in Eq. 3 as

$$Nu = \frac{hd}{k} \tag{3}$$

With k being the thermal conductivity of the fluid being evaluated. This can be rearranged to find the convective heat transfer coefficient, h_w,

$$h_w = \frac{Nu_w K_w}{d}$$

With K_w being the thermal conductivity of the water, found to be 0.6702 $\frac{\text{W}}{\text{m}\cdot\text{K}}$. Solving for h_w:

$$h_w = \frac{49.51(0.6702)}{0.020} = 1658.9988 \approx 1659 \frac{\text{W}}{\text{m}^2 \cdot \text{K}}$$

The next step was to find h_a, which comes can be found using shell-side heat transfer relationships. The Reynolds number for the shell-side is shown in Eq. 4:

$$Re_s = \frac{D_e G_s}{\mu_s} \tag{4}$$

Where d_e is the equivalent diameter in meters, G_s is the shell-side mass flow rate, and μ_s is the dynamic viscosity of the fluid in the shell.

From the given assumptions, Eq. 5 is

$$S_m \approx B(S_D - d) \tag{5}$$

Where S_m is the shell-side effective cross flow area in m^2, B is the baffle spacing, S_D is the tube-to-tube pitch in m, and d is the tube diameter in m. From the boundary conditions, it is known that the baffle spacing is taken to be 50% of the inner diameter of the shell-and-tube heat exchanger. It is also

known that the tube-to-tube pitch distance is taken to be 1.25 times a tubes outer diameter, which in this case is $1.25 \times 0.020 = 0.025$ m. The tube configuration was taken to be a staggered design with angles of 30°between tubes. An equation was found during research that relates the tube count and the tube bundle diameter [6]:

$$D_b = d_o(\frac{N_t}{K_1})^{\frac{1}{n_1}}$$

Where D_b is the bundle diameter in m, N_t is the number of tubes, and K_1 and n_1 are 0.249 and 2.207 for a shell-and-tube heat exchanger with triangular pitch and 2 passes. Inputting values:

$$D_b = 0.020(\frac{200}{0.249})^{\frac{1}{0.207}} = 0.414 \text{ m}$$

So there overall tube diameter is 0.414 m. To account for space between the tube bundle and the inside of the shell, the inner shell diameter is said to be 0.5 m. The required baffle spacing is then 50% of the inner diameter, so 0.25 m. Going back to Eq. 5,

$$S_m \approx 0.25(0.025 - 0.020) = 1.25 \times 10^{-3} \text{ m}^2$$

The shell-side mass flow rate is given as

$$G_s = \frac{\dot{m}_{brayton}}{S_m} = \frac{882.2}{1.25 \times 10^{-3}} = 7.0576 \times 10^5 \frac{\text{kg}}{\text{s} \cdot \text{m}^2}$$

The equivalent diameter of the shell-and-tube heat exchanger is given as

$$D_e = \frac{4(C_{p,d}S_D^2 - \frac{\pi d^2}{4})}{\pi d}$$

Where $C_{p,d}$ is 0.86 for a triangular pitch configuration, and d is the tube diameter. Solving for D_e,

$$D_e = \frac{4(0.86(0.025)^2 - \frac{\pi(0.020)^2}{4}}{\pi(0.020)} = 0.01422 \text{ m}$$

Going back to Eq. 4 to solve for the Reynolds number, with $\mu_s = 2.7203 \times 10^{-5}$ Pa · s,

$$Re_s = \frac{0.01422(7.0576 \times 10^5)}{2.7203 \times 10^{-5}} = 3.6888 \times 10^8$$

Which is turbulent flow. The Kern method, shown in Eq. 6 was then used for determining the Nusselt number, which has a useful range of $2 \times 10^3 \leq Re_s < 1 \times 10^6$. The shell-side Reynolds number falls out of this acceptable range, however as stated in Assumption #7, this range will be neglected. The Prandtl number for air was found to be 0.6985.

$$Nu_a = 0.36Re_s^{0.55}Pr^{1/3} = 0.36(3.6888 \times 10^8)^{0.55}(0.6985)^{1/3} = 1.6449 \times 10^4 \quad (6)$$

Using the definition of the Nusselt number, Eq. 3, the thermal conductivity of the air can be found, using the thermal conductivity of air, 0.04013 $\frac{\text{W}}{\text{m} \cdot \text{K}}$:

$$h_a = \frac{Nu_a k_a}{d} = \frac{8913.2664(0.04013)}{0.020} = 33004.78 \frac{\text{W}}{\text{m}^2 \cdot \text{K}}$$

The total thermal conductivity of the preheater, U, can be found by using a resistance approach, with the thin-walled pipe assumption neglecting resistance from the tube walls.

$$U = (\frac{1}{h_a} + \frac{1}{h_w})^{-1} = (\frac{1}{33004.78} + \frac{1}{1658.9988})^{-1} = 1579.6 \frac{\text{W}}{\text{m}^2 \cdot \text{K}}$$

Using Eq. 1, the total area can be isolated:

$$A = \frac{UA}{U} = \frac{31842.118}{1518.17} = 20.158 \text{ m}^2$$

The total area is defined in Eq. 7

$$A = N\pi dL \tag{7}$$

Solving for L,

$$L = \frac{A}{N\pi d} = \frac{20.158}{200\pi(0.020)} = \boxed{1.604 \text{ m}}$$

So the length of the tubes in the preheater should be 1.6691 m.

Superheater. The length of the tubes in the superheater was found using a log-mean temperature approach, shown in Eq. 8:

$$L_{SH} = \frac{Q_{SH}}{U_{SH}\pi d_i \Delta T_{lm} N} \tag{8}$$

Q_{SH} was found to be 2614.9×10^3 W, and d_i, the inner diameter of the tube, is 0.050 m. Eq. 9 is used to find ΔT_{lm}, the log-mean temperature of the tubes:

$$\Delta T_{lm} = \frac{\Delta T_2 - \Delta T_1}{ln(\frac{\Delta T_2}{\Delta T_1})} \tag{9}$$

For counter flow, $\Delta T_1 = T_4 - T_7$, and $\Delta T_2 = T_{4,b} - T_{sat}$. All temperatures are known, so Eq. 9 can be found:

$$\Delta T_{lm} = \frac{((T_4 - T_7) - (T_{4,b} - T_{sat}))}{ln(\frac{T_{4,b} - T_{sat}}{T_4 - T_7})} = \frac{((533.85 - 525.85) - (530.99 - 407.8))}{ln(\frac{(533.85 - 525.85)}{(530.99 - 407.8)})} = 42.128 \text{ K}$$

The next step is the find U_{SH}, the total thermal conductivity in the superheater, defined in Eq. 10 from a resistance approach:

$$U_{SH} = (\frac{1}{h_a} + \frac{1}{h_{steam}})^{-1} \tag{10}$$

The Reynolds number for the flow in the annulus is Eq. 11

$$Re = \frac{\rho u_m D_h}{\mu} \tag{11}$$

Where u_m is the mean velocity of the fluid, and D_h is the equivalent hydraulic diameter for a concentric tube annular flow. The dynamic viscosity μ was found to be 1.5887×10^{-5} Pa $\cdot$ s. The mean velocity

can be found in Eq. 12 by converting mass flow rate,

$$u_m = \frac{\dot{m}_w}{\rho_w A} \tag{12}$$

Where $\dot{m}_w$ is the mass flow rate of the water, ρ is the density of the water in kg/m^3, and A is the cross-sectional area of the flow. ρ was found at the average temperature and pressure of the water to be 1.4641 kg/m^3, and A was found using Eq. 13:

$$A = N(\frac{\pi(d_o^2 - d_i^2)}{4}) \tag{13}$$

Which is the overall area of flow for an annulus with an inner diameter, d_i, of 0.050 m, and an outer diameter, d_o, of 0.080 m. Solving for A:

$$A = 200(\frac{\pi(0.080^2 - 0.050^2)}{4}) = 0.6126 \text{ m}^2$$

The mean velocity can then be calculated using Eq. 12:

$$u_m = \frac{10.62}{(1.4641)(0.6126)} = 11.84 \text{ m/s}$$

The equation for equivalent hydraulic diameter for a concentric tube annular flow is given as Eq. 14,

$$D_h = d_o - d_i \tag{14}$$

Using Eq. 11, the Reynolds number of the steam is

$$Re = \frac{1.4641(11.84)(0.080 - 0.050)}{1.5887 \times 10^{-5}} = 32777.78$$

This is turbulent internal flow, so Eq. 2 can be utilized, with $n = 0.4$ for the steam being heated and the Prandtl number found using the average temperature of the fluid and P_6.

$$Nu = 0.023(32777.78)^{0.8}(0.9785)^{0.4} = 93.41$$

The convective heat transfer coefficient for the steam can then be found be rearranging Eq. 3, with the thermal conductivity of the steam found to be 0.03348 $\frac{\text{W}}{\text{m}\cdot\text{K}}$:

$$h_s = \frac{93.41(0.03348)}{0.080 - 0.030} = 104.261 \frac{\text{W}}{\text{m}^2 \cdot \text{K}}$$

The next step was to find h_a, the convective heat transfer coefficient of the air from the Brayton cycle. Eq. 15 is the Reynolds number for the air flow, using the converted mass flow rate:

$$Re = \frac{4(\frac{\dot{m}}{N})}{\pi d_i \mu_a} = \frac{4(\frac{882.2}{200})}{\pi(0.050)(2.812 \times 10^{-5})} = 3966705.64 \tag{15}$$

This is turbulent flow, so Eq. 2 can be used, with $n = 0.3$ because the air is being cooled.

$$Nu = 0.023(3966705.64)^{0.8}(0.6995)^{0.3} = 3925.62$$

Finally, Eq. 3 can be used to find h_a,

$$h_a = \frac{3925.62(0.04195)}{0.050} = 3293.77 \ \frac{\text{W}}{\text{m}^2 \cdot \text{K}}$$

U_{SH} can then be solved using Eq. 10,

$$U_{SH} = (\frac{1}{3293.77} + \frac{1}{104.261})^{-1} = 101.062 \ \frac{\text{W}}{\text{m}^2 \cdot \text{K}}$$

All the information is now gathered to solve Eq. 8:

$$L_{SH} = \frac{2614.9 \times 10^3}{101.062\pi(0.050)(41.128)(200)} = \boxed{19.58 \text{ m}}$$

So the length of the tubes in the superheater should be 19.58 m in length.

Evaporator. The final section of the heat exchanger is the evaporator section. The same formula for length as the superheater will be utilized, shown in Eq. 16,

$$L_{EV} = \frac{Q_{EV}}{U_{EV}\pi d_{EV}\Delta T_{lm,EV}} \tag{16}$$

Q_{EV} is given to be 22946.6 kW. The log-mean temperature difference for the evaporator, $\Delta T_{lm,EV}$, can be found similarly to the superheater using Eq. 9, with $\Delta T_1 = T_{4,b} - T_{sat}$, and $\Delta T_2 = T_{4,c} - T_{sat}$ for a heat exchanger in counterflow. Solving using known temperatures,

$$\Delta T_{lm,EV} = \frac{((505.89 - 407.8) - (530.99 - 407.8))}{ln(\frac{(505.89-407.8)}{(530.99-407.8)})} = 110.164 \text{ K}$$

To find U_{EV}, the assumption is used that for phase changes like boiling, the convective heat transfer coefficient is typically very high. Because of this, its contribution to thermal resistance is very small compared to the single-phase flow inside the evaporator tubes, so $U_{EV} \approx h_a$ in the evaporator. The Reynolds number for the air flow is given in Eq. 17,

$$Re = \frac{4(\frac{\dot{m}_a}{N})}{\pi d_i \mu_a} = \frac{4(\frac{882.2}{200})}{\pi(0.050)(2.7792 \times 10^{-5})} = 4041595.1 \tag{17}$$

This is turbulent flow, so Eq. 2 is once again used, with $n = 0.3$ because the air is being cooled.

$$Nu = 0.023(4041595.1)^{0.8}(0.69899)^{0.3} = 3983.93$$

Eq. 3 is then used to find the convection heat transfer coefficient of the air:

$$h_a = \frac{3983.93(0.04109)}{0.050} = 3274.13 \ \frac{\text{W}}{\text{m}^2 \cdot \text{K}} \approx U_{EV}$$

Using Eq. 16 to solve for the length,

$$L_{EV} = \frac{22946.6 \times 10^3}{3274.13\pi(0.050)(110.164)(200)} = \boxed{2.025 \text{ m}}$$

Results and Discussion

Because pool boiling is occuring in the evaporator section, it is necessary to confirm that the actual heat flux q'' is below the critical heat flux, q''_{CHF}. The tube area of the evaporator is given as:

$$A_{EV} = N\pi dL$$

Solving for the actual heat flux:

$$q'' = \frac{Q_{EV}}{A_{EV}} = \frac{Q_{EV}}{N\pi dL} = \frac{22946.6 \times 10^3}{200\pi(0.050)(2.025)} = 360.69 \ \frac{\text{kW}}{\text{m}^2}$$

The critical heat flux can be found by using Eq. 18,

$$q''_{CHF} = C_Z h_{fg} \rho_v [\frac{\sigma_{lv} g(\rho_l - \rho_v)}{\rho_v^2}]^{1/4} \tag{18}$$

Where C_Z is the Zuber constant, which for finite surfaces is 0.131. h_{fg} is the enthalpy difference between vapor and liquid water at P_6, which was found to be 2160.167 $\frac{\text{kJ}}{\text{kg}}$. ρ_v is the density of water vapor at P_6, found to be 1.7024 kg/m^3. σ_{lv} is the surface tension on the surface of the pipes, which was found at P_6 to be 0.0519 N/m. g is the gravity constant, 9.81 m/s^2. ρ_l is the density of liquid water at P_6, 930.85 kg/m^3. Plugging in these values,

$$q''_{CHF} = (0.131)(2160.167)(1.7024)[\frac{(0.0519)(9.81)(930.85 - 1.7024)}{(1.7024)^2}]^{1/4} = 1722.02 \ \frac{\text{kW}}{\text{m}^2}$$

Because $q'' < q''_{CHF}$, the actual pool boiling is less than the maximum heat flux, so the result is physically possible.

Final Answers.

(1) $T_4 = 533.85$ K
(2) $T_5 = 500$ K
(3) $T_6 = 308.49$ K
(4) $T_7 = 525.85$ K
(5) $\dot{m}_{brayton} = 882.2$ kg/s
(6) $\dot{m}_{rankine} = 10.62$ kg/s
(7) $T_{sat} = 407.8$ K
(8) $T_{4,c} = 505.89$ K
(9) $T_{4,b} = 530.99$ k
(10) $Q_{preheater} = 4437.03$ kW
(11) $Q_{evaporator} = 22946.6$ kW
(12) $Q_{superheater} = 2615.9$ kW
(13) Shell-and-tube pitch configuration: Triangular, 30°angle between tubes.
(14) Shell-and-tube pitch distance: 0.025 m
(15) Shell-and-tube baffle spacing: 0.25 m
(16) Shell-and-tube shell diameter: 0.5 m.

(17) $L_{preheater} = 1.604$ m
(18) $L_{superheater} = 19.58$ m
(19) $L_{evaporator} = 2.025$ m
(20) $q''_{evaporator} = 360.69$ kW/m^2
(21) $q''_{CHF} = 1722.02$ kW/m^2

Sanity Check. The superheater component requires a significantly greater tube length than the other two components. Refering to Figs. 3, 4, and 5, it can be seen that the superheater is required to raise the temperature of the water almost to the temperature of the air. The small outlet temperature difference is about 8 K, requiring a much larger heat transfer area in order to support the temperature change.

From research, the optimum tube length to shell diameter will usually fall within the range of 5 to 10 for a shell-and-tube heat exchanger [6]. Computing this ratio:

$$\frac{1.604}{0.5} = 3.208$$

While this falls short of the researched optimum, indicating that the heat exchanger is somewhat short compared to its shell diameter, it is still a valid design that falls close to expected values. For sanity's sake, it shows that the sizing performed during the analysis was done correctly, even if the design can be improved.

The heat transfer rates required in each section of the heat exchanger totaled 30 MW from the required temperature values of each section, indicating that the processes followed in each of the three design homework's aligned and that the expected capability of the overall heat exchanger was reasonable.

References

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[4] Shuman, D., n.d., "SHELL TUBE HEAT EXCHANGER DESIGN," Lawrence Berkeley National Laboratory, accessed 03/31/2026, https://www-eng.lbl.gov/~shuman/NEXT/MATERIALS&COMPONENTS/Pressure_vessels/shellandtube.pdf

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[6] Sinnott, R. K., 1999, *Coulson & Richardson's Chemical Engineering, Volume 6: Chemical Engineering Design*, 3rd ed., Butterworth-Heinemann.

Appendix: Python Code

```
import CoolProp.CoolProp as CP
import numpy as np

pi = np.pi
N = 200
mdot_w = 10.62
mdot_a = 882.2
d_pre = 0.020

# Hot Side:
T_4c = 505.89      # K
T_5 = 500          # K
T_havg = (T_4c+T_5)/2
cp_ac = CP.PropsSI('C', 'T', T_havg, 'P', 101325 , 'Air')
mu_ah = CP.PropsSI('V', 'T', T_havg, 'P', 101325, 'Air')
Pr_ah = CP.PropsSI('PRANDTL', 'T', T_havg, 'P', 101325, 'Air')
k_a = CP.PropsSI('L', 'T', T_havg, 'P', 101325, 'Air')

# Cold Side:
T_6 = 308.49       # K
T_sat = 407.8      # K
T_cavg = (308.49 + 407.8) / 2
P_w = 310e3
mu_wc = CP.PropsSI('V', 'T', T_cavg, 'P', P_w, 'Water')
cp_wc = CP.PropsSI('C', 'T', T_cavg, 'P', P_w, 'Water')
Pr_wc = CP.PropsSI('PRANDTL', 'T', T_cavg, 'P', P_w, 'Water')
k_wc = CP.PropsSI('L', 'T', T_cavg, 'P', P_w, 'Water')

C_c = mdot_w * cp_wc
C_h = mdot_a * cp_ac
C_min = min(C_c, C_h)
C_max = max(C_c, C_h)
C_r = C_min / C_max

Q_max = C_min * (T_4c - T_6)
Q = 4437e3
eff = Q / Q_max
E = ((2/eff) - (1 + C_r)) / np.sqrt(1 + C_r**2)
NTU = -(1 / np.sqrt(1 + C_r**2)) * np.log((E - 1) / (E + 1))
UA = C_min * NTU

Re = (4 * (mdot_w / N)) / (np.pi * 0.020 * mu_wc)
Nu = 0.023*(Re**0.8)*Pr_wc**0.4
hw = (Nu*k_wc)/d_pre

S_D = 0.025
B = 0.25
S_m = B*(S_D - d_pre)
G_s = mdot_a / S_m
D_e = (4 * (0.86 * S_D**2 - (pi * d_pre**2) / 4)) / (pi * d_pre)
Re_s = (D_e*G_s)/mu_ah
Nu_a = 0.36*(Re_s)**(0.55)*Pr_ah**(1/3)
ha = (Nu_a*k_a)/d_pre
U = (1/ha + 1/hw)**(-1)
```

```
A = UA/U
L = A / (N*pi*d_pre)
print( Preheater Tube Length =, L)

# --- SUPERHEATER --- #
Q_SH = 2615.9e3

d_i = 0.050 # m
d_o = 0.080 # m
d_h = d_o - d_i

T_4b = 530.99 # T_h,o [K]
T_4 = 533.85 # T_h,i [K]
T_havg_SH = (T_4b+T_4)/2

T_7 = 525.85 # T_c,o [K]
T_cavg_SH = (T_7+T_sat)/2

dT1 = T_4 - T_7
dT2 = T_4b - T_sat

dT_lm = (dT2-dT1)/(np.log((dT2/dT1)))

mu_a_SH = CP.PropsSI('V', 'T', T_havg_SH, 'P', 101325, 'Air')
Re_a = (4*(mdot_a/N))/(pi*d_i*mu_a_SH) # Turbulent
Pr_a_SH = CP.PropsSI('PRANDTL', 'T', T_havg_SH, 'P', 101325, 'Air')
k_a_SH = CP.PropsSI('L', 'T', T_havg_SH, 'P', 101325, 'Air')
Nu_a = (0.023*(Re_a)**0.8)*Pr_a_SH**0.3
h_a = (Nu_a*k_a_SH)/d_i

rho_s = CP.PropsSI('D', 'T', T_cavg_SH, 'P', P_w, 'Water')
mu_w_SH = CP.PropsSI('V', 'T', T_cavg_SH, 'P', P_w, 'Water')
Pr_w_SH = CP.PropsSI('PRANDTL', 'T', T_cavg_SH, 'P', P_w, 'Water')
k_steam = CP.PropsSI('L', 'T', T_cavg_SH, 'P', P_w, 'Water')
A_ann = N*(pi/4)*(d_o**2-d_i**2)
u_m = (mdot_w / (rho_s*A_ann))

Re_steam = (rho_s*u_m*d_h)/mu_w_SH
Nu_steam = 0.023*(Re_steam**0.8)*Pr_w_SH**0.4
h_steam = (Nu_steam*k_steam)/d_h
U_SH = (1/h_a + 1/h_steam)**-1
L_SH = Q_SH/(U_SH*pi*d_i*dT_lm*N)
print( Superheater Tube Length =, L_SH)


# --- EVAPORATOR --- #
d_EV = 0.050
Q_EV = 22946.6e3

T_havg_EV = (T_4b+T_4c)/2
dT1_EV = T_4b - T_sat
dT2_EV = T_4c - T_sat
dT_lm_EV = (dT1_EV - dT2_EV) / np.log(dT1_EV / dT2_EV)

mu_a_EV = CP.PropsSI('V', 'T', T_havg_EV, 'P', 101325, 'Air')
Pr_gas = CP.PropsSI('PRANDTL', 'T', T_havg_EV, 'P', 101325, 'Air')
k_gas = CP.PropsSI('L', 'T', T_havg_EV, 'P', 101325, 'Air')
```

```
112
113 Re_gas = (4*(mdot_a/N))/(pi*d_EV*mu_a_EV) # Turbulent
114 Nu_gas = 0.023*(Re_gas**0.8)*Pr_gas**0.3
115 h_gas = (Nu_gas*k_gas)/d_EV
116
117 U_EV = (1/h_gas)**(-1)
118 L_EV = Q_EV/(U_EV*pi*d_EV*dT_lm_EV*N)
119 print( Evaporator Tube Length =, L_EV)
120
121 A_EV = N*pi*d_EV*L_EV
122 flux_EV = Q_EV/A_EV
123
124 g = 9.81
125 C_Z = 0.131
126 h_f = CP.PropsSI('H', 'P', P_w, 'Q', 0, 'Water')
127 h_g = CP.PropsSI('H', 'P', P_w, 'Q', 1, 'Water')
128 h_fg = h_g - h_f
129 rho_l = CP.PropsSI('D', 'P', P_w, 'Q', 0, 'Water')
130 rho_v = CP.PropsSI('D', 'P', P_w, 'Q', 1, 'Water')
131 sigma_lv = CP.PropsSI('I', 'P', P_w, 'Q', 0, 'Water')
132
133 q_CHF = C_Z*h_fg*rho_v*((sigma_lv*g*(rho_l-rho_v))/rho_v**2)**0.25
```